

AutoSentinel— A Dynamic Driver Fatigue State and Fleet Monitoring System

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Abstract—Driver fatigue is a major contributor to road accidents, particularly during prolonged driving and commercial fleet operations. This paper presents AutoSentinel, a real-time driver monitoring and fleet management system combining computer vision, personalized behavioral analysis, physical alerting, and cloud monitoring. A driver-facing camera with MediaPipe FaceMesh extracts facial landmarks to compute Eye Aspect Ratio (EAR), Percentage of Eye Closure (PERCLOS), Mouth Aspect Ratio (MAR), yawning activity, eye closure duration, and head-pose characteristics.

To account for inter-driver variability, the system performs a 60-second calibration during registration to establish an individual baseline for subsequent monitoring. The extracted features are processed to classify the driver as Normal, Drowsy, or Unresponsive. For critical states, an Arduino-based alert subsystem provides visual and audible warnings, while a hardware lockout requires physical acknowledgement before system reset. Driver telemetry and fatigue events are stored in a cloud backend and presented through a centralized fleet dashboard for real-time supervision and historical analysis.

Index Terms—Driver Fatigue Detection, Computer Vision, Eye Aspect Ratio (EAR), PERCLOS, Facial Landmark Detection, MediaPipe FaceMesh, Real-Time Systems, Fleet Monitoring, Edge-Cloud Architecture, Drowsiness Detection

I. INTRODUCTION

Driver fatigue is a major road-safety concern, particularly during prolonged driving and commercial fleet operations. Observable indicators such as eye closure, blinking, yawning, mouth opening, and head posture support non-invasive computer-vision-based monitoring [1]. Recent studies emphasize combining multiple behavioral indicators, with PERCLOS being a widely investigated measure of drowsiness [2], [3], [4], [5], [6], [7].

Inter-driver variability in facial geometry and habitual behavior can make uniform thresholds unsuitable. AutoSentinel addresses this through a **60-second personalized calibration** during registration to establish an individual baseline.

The system uses **MediaPipe FaceMesh** to extract EAR, PERCLOS, MAR, yawning activity, eye-closure duration, head pose, and nose position. These ocular, oral, and head-related features support multi-factor driver-state assessment [8], [9]. The system classifies drivers as **Normal, Drowsy, or Unresponsive**.

AutoSentinel further integrates an **Arduino-based alert subsystem** with visual and audible indicators, physical acknowledgement, and a hardware lock for Unresponsive conditions [10], [11]. Cloud connectivity enables centralized storage of driver telemetry and fatigue events.

The main contributions are:

- **60-second personalized calibration** for individual driver baselines.
- **Multi-factor facial analysis** using ocular, oral, and head-related indicators.
- **Arduino-based physical alert and safety interlock** for critical states.
- **Cloud-based fleet monitoring** and historical event analysis.

II. LITERATURE SURVEY

Driver fatigue detection has been studied using computer vision, machine learning, physiological sensing, and multimodal approaches. Recent work emphasizes non-contact facial and

behavioral monitoring, including OpenCV- and MediaPipe-based driver health monitoring [12], [2], [3], [4].

EAR, PERCLOS, and blink behavior are widely used for fatigue assessment. PERCLOS measures prolonged eye closure, while blink characteristics provide complementary information [6], [7], [13]. However, individual indicators may not capture all fatigue-related behavioral changes.

Multimodal methods combine complementary indicators. Du et al. combined heart rate with PERCLOS, while Bai et al. investigated facial-feature-based fatigue detection. Facial and skeletal features have also been explored for driver-state representation [8], [9], [14], [15].

Inter-driver variability in facial geometry and habitual head position remains a challenge. The reviewed references do not directly establish AutoSentinel's specific 60-second calibration approach; therefore, the system introduces driver-specific calibration to establish an individual baseline.

Commercial and fleet systems provide continuous monitoring, alerts, and driver analytics, with some incorporating steering and lane-deviation information [16], [17], [18], [19]. Predictive and explainable fatigue assessment has also been investigated [20], [21].

AutoSentinel extends these approaches by combining **personalized calibration, multi-factor facial analysis, Arduino-based physical intervention, and cloud-based fleet monitoring**.

III. SYSTEM ARCHITECTURE

The AutoSentinel architecture integrates **in-cabin perception, personalized calibration, fatigue analysis, physical intervention, and cloud-based fleet monitoring** [14]. The overall architecture is shown in Fig. 1.

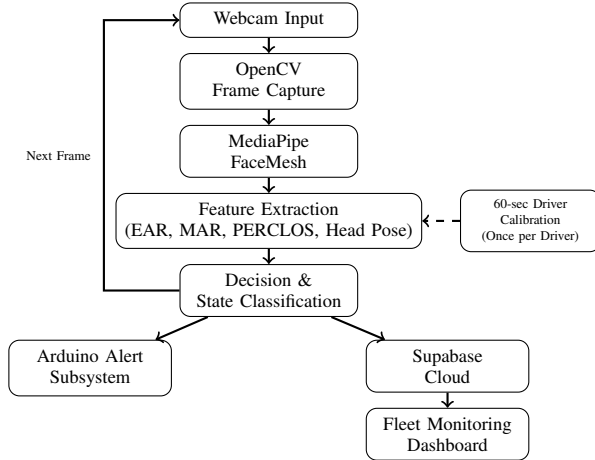


Fig. 1. System architecture of the proposed AutoSentinel system with one-time driver calibration and continuous real-time processing.

The architecture supports continuous driver monitoring, intervention, and fleet supervision.

A. Vision Input Module

The webcam captures driver video through OpenCV for continuous analysis.

B. Facial Landmark Detection Module

MediaPipe FaceMesh extracts facial landmarks for ocular, oral, and head-related feature analysis [22].

C. Driver Calibration Module

A **60-second calibration** establishes an individual baseline for eye openness, mouth configuration, and head position.

D. Fatigue Analysis Module

The system computes EAR, MAR, PERCLOS, blink count, yawning, eye-closure duration, head pose, and nose position.

E. Decision Engine

Personalized measurements classify the driver as **Normal, Drowsy, or Unresponsive** and forward the state to the hardware and cloud layers [11].

F. Hardware Alert Module

Arduino controls LEDs, buzzer, and physical acknowledgment, with a **hardware lock for Unresponsive conditions**.

G. Cloud Database Module

Supabase stores driver-session data, fatigue metrics, timestamps, states, and events.

H. Fleet Monitoring Dashboard

The **ReactJS dashboard** provides centralized visualization of driver states, fatigue metrics, trends, and alerts [23].

IV. METHODOLOGY

The AutoSentinel methodology integrates **computer vision, personalized calibration, multi-factor fatigue analysis, physical alerts, and cloud-based fleet monitoring**.

A. Vision Input Module

1) *Overview*: The module captures real-time driver video and preprocesses frames for analysis.

2) *Frame Acquisition and Preprocessing*: Video is acquired using OpenCV's VideoCapture interface [24]. Each frame undergoes:

- Horizontal flipping for mirror correction
- BGR-to-RGB conversion:

$$Frame_{RGB} = Convert(Frame_{BGR}) \quad (1)$$

- CLAHE-based illumination normalization
- Frame timestamping

B. Face Detection and Landmark Extraction

MediaPipe Face Detection and FaceMesh extract 468 normalized 3D facial landmarks [22]:

$$L = \{(x_i, y_i, z_i)\}_{i=1}^{468} \quad (2)$$

The system is configured for single-driver tracking with refined landmarks.

C. Personalized Driver Calibration

AutoSentinel performs a **60-second personalized calibration** during registration to establish an individual baseline for eye openness, mouth configuration, and head position.

Subsequent measurements are interpreted relative to this baseline, reducing errors caused by **inter-driver differences in facial characteristics and resting behavior**.

D. Eye Landmark Extraction

From the facial landmark set, key eye landmarks are extracted:

Left Eye: [33, 160, 158, 133, 153, 144]

Right Eye: [362, 385, 387, 263, 373, 380]

These points represent eyelid geometry and are used for eye behavior analysis.

E. Fatigue Feature Extraction

The system extracts **ocular, oral, and head-related indicators** from facial landmarks.

1) *Adaptive Eye Aspect Ratio (EAR)*: The Eye Aspect Ratio (EAR) provides a frame-level measure of eye openness:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \cdot \|p_1 - p_4\|} \quad (3)$$

During the 60-second calibration stage, EAR observations are collected to establish a personalized baseline:

$$EAR_{\text{base}} = \text{median}(EAR_1, EAR_2, \dots, EAR_N) \quad (4)$$

During monitoring, a rolling buffer of 30 frames is used to reduce frame-level fluctuations:

$$EAR_{\text{med}}(t) = \text{median}(EAR_{t-29}, \dots, EAR_t) \quad (5)$$

The current eye-openness condition is evaluated relative to the personalized baseline:

$$R_{EAR}(t) = \frac{EAR_{\text{med}}(t)}{EAR_{\text{base}}} \quad (6)$$

This adaptive comparison reduces dependence on a single population-level EAR threshold and provides more consistent interpretation across drivers with different facial geometry and natural eye openness.

2) *Mouth Aspect Ratio and Yawning*: **MAR** quantifies mouth opening from facial landmarks. Sustained increases identify **yawning events**, which are used for real-time fatigue analysis and longitudinal scoring.

3) *Head Pose and Nose Position*: Head pose and nose position track **head orientation, tilt, and positional changes**, complementing ocular and oral indicators for monitoring driver posture and attention.

F. Composite Fatigue Score

Multiple features are combined into a normalized fatigue score:

$$FS = w_1 \cdot EAR + w_2 \cdot PERCLOS + w_3 \cdot BR + w_4 \cdot ECD \quad (7)$$

Default weights:

$$(w_1, w_2, w_3, w_4) = (0.25, 0.40, 0.15, 0.20)$$

The score is scaled to **0–100** and temporally smoothed. MAR, yawning, head pose, and nose position provide additional decision information.

G. Driver State Classification

Driver states are classified using the EAR comparison, PERCLOS, and temporal persistence criteria shown in Table I.

TABLE I
DRIVER STATE CLASSIFICATION AND TRIGGERED ACTIONS

State	EAR Comparison	PERCLOS	Action
Normal	30-frame median near baseline	< 0.35	None
Drowsy	30-frame median reduced from baseline	0.35–0.70	Audio alert + cloud update
Unresponsive	Sustained reduction in 30-frame median	> 0.70	Continuous alarm + emergency flag

To reduce frame-level noise, EAR values are evaluated using a rolling 30-frame buffer. The median EAR is compared with the driver's calibrated baseline, reducing the effect of transient landmark errors. Driver state is determined from this comparison together with PERCLOS and condition persistence.

The ratio R_{EAR} represents current eye openness relative to the individual baseline. Values near the baseline indicate normal openness, while sustained reductions indicate increased eye closure. This ratio, PERCLOS, and temporal persistence are jointly considered for classification.

H. Hardware Alert and Safety Interlock

The classified state is transmitted to the Arduino through a Python USB serial bridge.

The **Normal** state activates the green LED, while **Drowsy** activates intermittent red LED and buzzer warnings. **Unresponsive** activates continuous alerts and a hardware lock.

A **physical push button** acknowledges the critical state, followed by stabilization and reset before returning to Normal.

I. Longitudinal Driver Scoring Architecture

AutoSentinel converts instantaneous measurements into **time-normalized historical rates and composite performance indices**.

1) *Historical Parameter Derivation*: Telemetry events are normalized across cumulative driving hours (T_{drive}):

- **Yawn Rate ($YR, LongYR$):**

$$YR = \frac{\text{Total Yawns}}{T_{\text{drive}}} \quad (8)$$

$$LongYR = \frac{\text{Yawns} > \tau_{\text{yawn}}}{T_{\text{drive}}} \quad (9)$$

- **Microsleep Exposure (MR, D_{max}):**

$$MR = \frac{\text{Microsleep Count}}{T_{\text{drive}}}, \quad D_{\text{max}} = \max_i(ECD_i) \quad (10)$$

- **High-PERCLOS Percentage ($PERCLOS_{\text{high}}$):**

$$PERCLOS_{\text{high}} = \left(\frac{\text{Mins with } PERCLOS > \theta_{\text{perclos}}}{T_{\text{drive}} \cdot 60} \right) \times 100 \quad (11)$$

- **Distraction Rate (DR):** Measures sustained head pose deviations (> 2.5 s):

$$DR = \frac{\text{Sustained Deviations}}{T_{\text{drive}}} \quad (12)$$

2) *Composite Driver Score (DS) and Capping*: The Driver Score ($DS \in [0, 100]$) integrates fatigue resilience (F), safety history (S), attention (A), experience (E), and medical fitness (H):

$$DS = 0.35F + 0.25S + 0.15A + 0.15E + 0.10H \quad (13)$$

Score caps prevent severe fatigue events from being obscured by other components:

$$DS_{\text{final}} = \begin{cases} \min(DS, 75), & \text{if 1 unresponsive event} \\ \min(DS, 50), & \text{if } > 1 \text{ unresponsive events} \\ DS, & \text{otherwise} \end{cases} \quad (14)$$

3) *Dynamic Route Suitability Score (RSS)*: The **Dynamic Route Suitability Score (RSS)** evaluates 7-day fatigue trends, shift hours, and route duration to determine immediate driver readiness independently of DS_{final} .

J. Cloud Integration and Data Pipeline

Relevant driver events are uploaded to Supabase, including:

- Driver ID and timestamp
- EAR, PERCLOS, BR, and ECD
- MAR and yawning events
- Head pose and nose position
- Fatigue Score and Driver State
- Captured event frames

Asynchronous communication prevents cloud operations from interrupting local monitoring and physical alerts.

K. Fleet Monitoring Dashboard

A **ReactJS-based dashboard** provides centralized fleet monitoring through:

- Live driver feed and fatigue metrics
- Driver state visualization
- Time-series fatigue trends
- Alert logs with captured frames
- Historical driver performance

The dashboard periodically refreshes to support **fleet-level supervision of driver states, fatigue events, and longitudinal trends**.

V. TECHNOLOGY STACK

AutoSentinel integrates computer vision, data processing, embedded hardware, cloud storage, and ReactJS-based fleet visualization.

TABLE II
TECHNOLOGY STACK OF AUTOSENTINEL

Component	Technology
Programming Language	Python
Computer Vision	OpenCV [24]
Facial Landmark Detection	MediaPipe FaceMesh [22]
Data Processing	NumPy, Pandas
Fatigue Analysis	EAR, MAR, PERCLOS, Blink Rate, ECD, Yawning, Head Pose, Nose Position
Hardware Controller	Arduino Uno
Communication	USB Serial
Alert Hardware	Red/Green LEDs, Active Buzzer, Push Button
Cloud Database	Supabase [25]
Dashboard	ReactJS
Data Visualization	Matplotlib

VI. RESULTS

A. Driver State Detection

AutoSentinel was evaluated using real-time facial video under different driver conditions. The implemented system identified **Normal**, **Drowsy**, and **Unresponsive** states while displaying the corresponding facial and fatigue-related measurements.

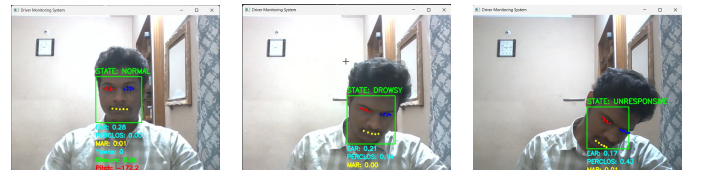


Fig. 2. Real-time driver-state detection: (a) Normal, (b) Drowsy, and (c) Unresponsive.

The Normal-state observation shows an **EAR of 0.28, PERCLOS of 0.00, MAR of 0.01**, zero detected yawns, and zero seconds of eye closure. The Drowsy-state observation shows an EAR of 0.21 and PERCLOS of 0.14. The Unresponsive-state observation demonstrates continued facial tracking and driver-state classification.

These results demonstrate **real-time facial landmark tracking and multi-feature fatigue measurement** for driver-state monitoring.

B. Hardware Alert System

The driver-state classifier was integrated with an **Arduino-based physical alert subsystem** consisting of an Arduino Uno, LEDs, active buzzer, and physical push button.

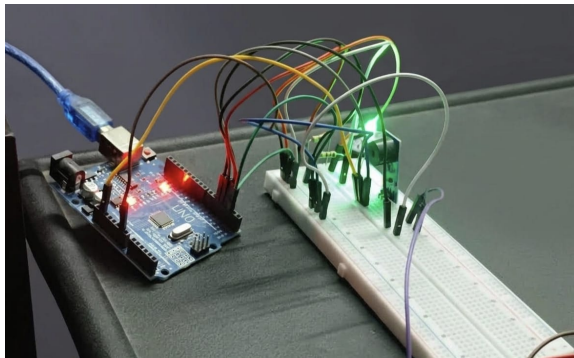


Fig. 3. Arduino-based physical driver alert and acknowledgement subsystem.

The Arduino receives the classified state through serial communication. The **Normal state activates the green LED**, while the **Drowsy state activates intermittent red LED and buzzer warnings**. The **Unresponsive state activates a continuous critical alert and hardware lock**. A physical push button is required to acknowledge the critical condition before reset.

This demonstrates physical intervention in addition to software-based fatigue detection.

C. Fleet Monitoring Dashboard

The implemented **ReactJS-based dashboard** provides centralized fleet supervision through interconnected fleet and driver management views.



Fig. 4. AutoSentinel fleet dashboard overview.

The overview displays **total fleets, total drivers, active drivers, and alerts**, with navigation to fleet, driver, alert, and profile modules.

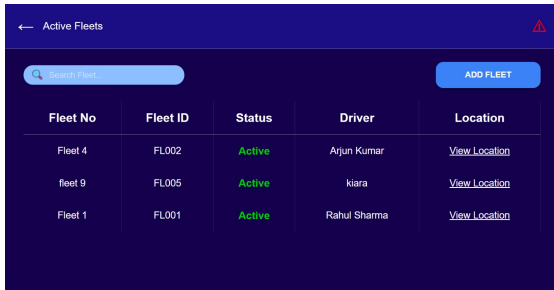


Fig. 5. Active Fleets view showing fleet status, assigned drivers, and locations.

The Active Fleets view provides **fleet ID, status, assigned driver, and location**, with search and fleet creation functionality.

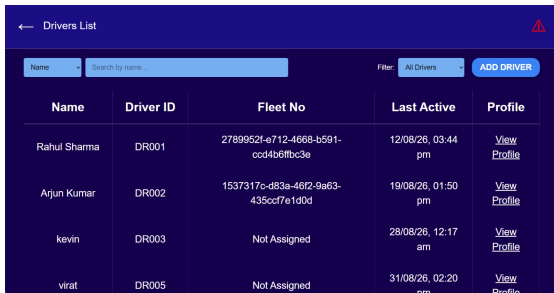


Fig. 6. Drivers List interface for driver search, filtering, and profile access.

The Drivers List provides **driver ID, fleet information, last active time, and profile access**, with search and filtering.

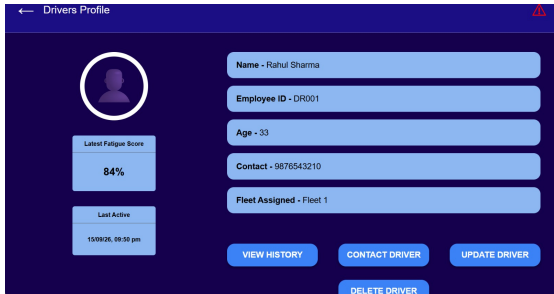


Fig. 7. Individual driver profile displaying driver information and fatigue status.

The Driver Profile displays **driver details, fleet assignment, fatigue score, and last active time**, with history and driver-management options.



Fig. 8. Individual driver history and fatigue analysis view.

The analysis view presents **average and maximum fatigue levels, drowsy and unresponsive events, and Driver Safety Score history** for longitudinal analysis.

VII. ADVANTAGES

- **Non-Intrusive:** Camera-based real-time monitoring without wearable sensors.
- **Personalized:** 60-second calibration establishes an individual baseline.
- **Multi-Factor:** Combines EAR, PERCLOS, MAR, blinking, yawning, ECD, head pose, and nose position.
- **Physical Intervention:** Arduino LEDs, buzzer, acknowledgement, and hardware lock provide driver-facing alerts.
- **Centralized Monitoring:** Cloud storage and dashboard provide driver states, fatigue metrics, alerts, and history.
- **Modular:** Separates vision processing, hardware, cloud storage, and fleet visualization.

VIII. LIMITATIONS

- **Lighting:** Extreme lighting, glare, or low illumination may affect landmark detection.
- **Face Visibility:** Sunglasses, masks, occlusion, or poor camera placement may reduce reliability.
- **Camera Dependency:** Accurate monitoring requires a properly positioned driver-facing camera.
- **Internet Dependency:** Cloud synchronization and dashboard updates require connectivity; local detection and alerts remain independent.
- **Behavioral Scope:** Facial and head indicators may not capture fatigue without observable behavioral changes.
- **Calibration Dependency:** The baseline depends on representative calibration conditions and may change with camera or driving conditions.

IX. FUTURE SCOPE

- **Edge Deployment:** Deploy on Raspberry Pi or Jetson-class platforms.
- **Adaptive Models:** Incorporate temporal machine learning for personalized fatigue patterns.
- **Vehicle Integration:** Integrate driver-assistance systems for controlled intervention and safe-stop mechanisms [26], [27], [28].
- **Offline Communication:** Add GPS and cellular communication for tracking and event transmission [29].
- **Predictive Analytics:** Analyze historical data for fatigue trends, risk profiling, and route or shift planning.

X. CONCLUSION

AutoSentinel combines **MediaPipe FaceMesh, personalized calibration, multi-factor fatigue analysis, Arduino-based physical intervention, cloud storage, and ReactJS fleet monitoring**. The system classifies drivers as **Normal, Drowsy, or Unresponsive** and provides visual, audible, and hardware-locked intervention for critical conditions.

The implementation demonstrates the feasibility of integrating **real-time driver monitoring, physical intervention,**

and centralized fleet supervision. Future work includes edge deployment, adaptive AI, offline communication, vehicle integration, and predictive analytics.

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